

## WEEK 2 – IDENTIFYING DATA REQUIREMENTS

### DATABASE REQUIREMENT ANALYSIS

#### PROJECT TITLE: E-COMMERCE SHOPPING PLATFORM

##### 1. INTRODUCTION

In Week 2, the data requirements of the proposed e-commerce shopping platform are identified.

The database is required to store information about customers, products, categories, suppliers, orders, payments, shipments, and product reviews.

Each entity contains different attributes, and every entity has a primary key that uniquely identifies each record.

##### 2. IDENTIFIED ENTITIES

The main entities identified for the e-commerce system are:

1. Customer
2. Product
3. Category
4. Supplier
5. Order
6. Payment
7. Shipment
8. Review

##### 9. ENTITY AND ATTRIBUTE DETAILS

#### 3.1 CUSTOMER

Purpose:

Stores information about customers who use the e-commerce platform.

Attributes:

- Customer\_ID – Unique identification number of the customer.
- First\_Name – Customer's first name.
- Last\_Name – Customer's last name.
- Email – Customer's email address.
- Phone – Customer's phone number.
- Password – Customer's login password.
- Address – Customer's delivery address.
- City – Customer's city.
- State – Customer's state.
- Pincode – Customer's postal code.

Primary Key:  
Customer\_ID

### 3.2 PRODUCT

Purpose:  
Stores information about products available for sale.

Attributes:

- Product\_ID – Unique identification number of the product.
- Product\_Name – Name of the product.
- Description – Description of the product.
- Price – Price of the product.
- Stock\_Quantity – Available quantity of the product.
- Brand – Brand name of the product.
- Category\_ID – Identifies the category of the product.
- Supplier\_ID – Identifies the supplier of the product.

Primary Key:  
Product\_ID

### 3.3 CATEGORY

Purpose:  
Stores information about product categories.

Attributes:

- Category\_ID – Unique identification number of the category.
- Category\_Name – Name of the category.
- Description – Description of the category.

Primary Key:  
Category\_ID

### 3.4 SUPPLIER

Purpose:  
Stores information about suppliers who provide products.

Attributes:

- Supplier\_ID – Unique identification number of the supplier.
- Supplier\_Name – Name of the supplier.
- Email – Supplier's email address.
- Phone – Supplier's phone number.
- Address – Supplier's address.
- City – Supplier's city.

- State – Supplier's state.
- Pincode – Supplier's postal code.

Primary Key:  
Supplier\_ID

### 3.5 ORDER

Purpose:  
Stores information about customer orders.

Attributes:

- Order\_ID – Unique identification number of the order.
- Customer\_ID – Identifies the customer who placed the order.
- Order\_Date – Date on which the order was placed.
- Total\_Amount – Total amount of the order.
- Order\_Status – Current status of the order.
- Delivery\_Address – Address where the order should be delivered.

Primary Key:  
Order\_ID

### 3.6 PAYMENT

Purpose:  
Stores information about payments made for orders.

Attributes:

- Payment\_ID – Unique identification number of the payment.
- Order\_ID – Identifies the order associated with the payment.
- Payment\_Date – Date of payment.
- Payment\_Method – Method used for payment.
- Payment\_Amount – Amount paid.
- Payment\_Status – Current status of the payment.
- Transaction\_ID – Unique transaction reference.

Primary Key:  
Payment\_ID

### 3.7 SHIPMENT

Purpose:  
Stores information about the shipment and delivery of orders.

Attributes:

- Shipment\_ID – Unique identification number of the shipment.
- Order\_ID – Identifies the order being shipped.

- Shipment\_Date – Date when the order was shipped.
- Delivery\_Date – Expected or actual delivery date.
- Tracking\_Number – Unique tracking number.
- Shipment\_Status – Current shipment status.
- Delivery\_Address – Address where the shipment will be delivered.

Primary Key:  
Shipment\_ID

### 3.8 REVIEW

Purpose:  
Stores customer reviews and ratings for products.

Attributes:

- Review\_ID – Unique identification number of the review.
- Customer\_ID – Identifies the customer who submitted the review.
- Product\_ID – Identifies the product being reviewed.
- Rating – Rating given by the customer.
- Comment – Customer's review or feedback.
- Review\_Date – Date when the review was submitted.

Primary Key:  
Review\_ID

## 4. SUMMARY OF ENTITIES AND PRIMARY KEYS

### Entity Primary Key

Customer Customer\_ID  
Product Product\_ID  
Category Category\_ID  
Supplier Supplier\_ID  
Order Order\_ID  
Payment Payment\_ID  
Shipment Shipment\_ID  
Review Review\_ID

## 5. FOREIGN KEY ATTRIBUTES

Some entities are connected to other entities using foreign keys.

- Product → Category\_ID
- Product → Supplier\_ID
- Order → Customer\_ID
- Payment → Order\_ID
- Shipment → Order\_ID

- Review → Customer\_ID
- Review → Product\_ID

These foreign keys help establish relationships between different entities in the database.

## 6. DATA REQUIREMENT SUMMARY

The database will store the following major types of information:

### CUSTOMER DATA:

Customer name, email, phone number, password, and address.

### PRODUCT DATA:

Product name, description, price, brand, stock quantity, category, and supplier.

### CATEGORY DATA:

Category name and category description.

### SUPPLIER DATA:

Supplier name, contact details, and address.

### ORDER DATA:

Order details, customer information, order date, amount, status, and delivery address.

### PAYMENT DATA:

Payment method, amount, transaction ID, payment date, and payment status.

### SHIPMENT DATA:

Shipment date, delivery date, tracking number, shipment status, and delivery address.

### REVIEW DATA:

Customer rating, comments, product information, and review date.

## 7. CONCLUSION

The identified entities and attributes provide the basic data requirements for the proposed e-commerce shopping platform.

Each entity has a primary key that uniquely identifies its records. Foreign keys are used to establish relationships between entities.

This data requirement analysis will be useful for designing the database schema and Entity Relationship Diagram (ER Diagram) in the next stages of the project.